

## Chapter-2

# BOOLEAN ALGEBRA

### ➤ Introduction:

- An algebra that deals with binary number system is called “Boolean Algebra”.
- It is very power in designing logic circuits used by the processor of computer system.
- The logic gates are the building blocks of all the circuit in a computer.
- Boolean algebra derives its name from the mathematician **George Boole** (1815-1864) who is considered the “**Father of symbolic logic**”.
- Boolean algebra deals with truth table TRUE and FALSE.
- It is also called as “**Switching Algebra**”.

### ➤ Binary Valued Quantities – Variable and Constants:

- A variable used in Boolean algebra or Boolean equation can have only one of two variables. The two values are FALSE (0) and TRUE (1)
- A Sentence which can be determined to be TRUE or FALSE are called **logical statements** or **truth functions** and the results TRUE or FALSE is called **Truth values**.
- The variables which can store the truth values are called **logical variables or binary valued variables**. These can store one of the two values 1 or 0.
- The decision which results into either YES (TRUE or 1) or NO (FALSE or 0) is called **Binary decision**.

### ➤ Truth Table:

- A truth table is a mathematical table used in logic to computer functional values of logical expressions.
- A truth table is a table whose columns are statements and whose rows are possible scenarios.
- Example: Consider the logical expression

Logical Statement: Meals = “Ram prefer rice and roti for the meal”

**Y = A AND B** (Logical Variables: Y, A, B, Logical Operator AND)

| Ram Prefer Rice | Ram Prefer Roti | Meals |
|-----------------|-----------------|-------|
| FALSE           | FALSE           | FALSE |
| FALSE           | TRUE            | FALSE |
| TRUE            | FALSE           | FALSE |
| TRUE            | TRUE            | TRUE  |

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

- If result of any logical statement or expression is always TRUE or 1, it is called **Tautology** and if the result is always FALSE or 0, it is called **Fallacy**.

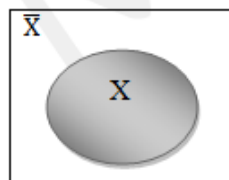
### ➤ Logical Operators:

- There are three logical operator, NOT, OR and AND.
- These operators are now used in computer construction known as switching circuits.

### ➤ NOT Operator:

- The Not operator is a unary operator. This operator operates on single variable.
- The operation performed by Not operator is called **complementation**.
- The symbol we use for it is bar.
- $\bar{X}$  means complementation of X
- If  $X=1$ ,  $\bar{X}=0$       If  $X=0$ ,  $\bar{X}=1$
- The Truth table and the Venn diagram for the NOT operator is:

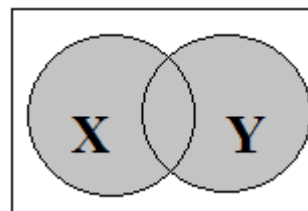
| X | $\bar{X}$ |
|---|-----------|
| 1 | 0         |
| 0 | 1         |



### ➤ OR Operator:

- The OR operator is a binary operator. This operator operates on two variables.
- The operation performed by OR operator is called **logical addition**.
- The symbol we use for it is '+'. Example:  $X + Y$  can be read as **X OR Y**
- The Truth table and the Venn diagram for the NOT operator is:

| X | Y | $X + Y$ |
|---|---|---------|
| 0 | 0 | 0       |
| 0 | 1 | 1       |
| 1 | 0 | 1       |
| 1 | 1 | 1       |

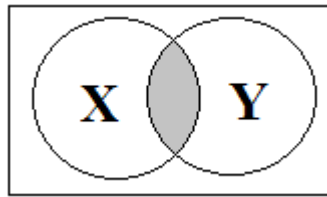


### ➤ AND Operator:

- The AND operator is a binary operator. This operator operates on two variables.
- The operation performed by AND operator is called **logical multiplication**.
- The symbol we use for it is '·'.
- Example:  $X \cdot Y$  can be read as **X AND Y**

- The Truth table and the Venn diagram for the NOT operator is:

| X | Y | $X \cdot Y$ |
|---|---|-------------|
| 0 | 0 | 0           |
| 0 | 1 | 0           |
| 1 | 0 | 0           |
| 1 | 1 | 1           |



### ➤ Evaluation of Boolean Expression using Truth Table:

- To create a truth table, follow the steps given below.**
- Step 1: Determine the number of variables, for  $n$  variables create a table with  $2^n$  rows.
  - For two variables i.e.  $X, Y$  then truth table will need  $2^2$  or 4 rows.
  - For three variables i.e.  $X, Y, Z$ , then truth table will need  $2^3$  or 8 rows.
- Step 2: List the variables and every combination of 1 (TRUE) and 0 (FALSE) for the given variables
- Step 3: Create a new column for each term of the statement or argument.
- Step 4: If two statements have the same truth values, then they are equivalent.

### ➤ Example: Consider the following Boolean Expression $F = X + \bar{Y}$

- Step 1: This expression as two variables  $X$  and  $Y$ , then  $2^2$  or 4 rows.
- Step 2: List the variables and every combination of  $X$  and  $Y$ .
- Step 3: Create a new column  $\bar{Y}$  of the statement, and then fill the truth values of  $Y$  in that column.
- Step 4: The final column contain the values of  $X + \bar{Y}$ .

| X | Y | $\bar{Y}$ | $X + \bar{Y}$ |
|---|---|-----------|---------------|
| 0 | 0 | 1         | 1             |
| 0 | 1 | 0         | 0             |
| 1 | 0 | 1         | 1             |
| 1 | 1 | 0         | 1             |

### ➤ Exercise Problems:

- Prepare a table of combination for the following Boolean algebra expressions.
  - $\bar{X}\bar{Y} + \bar{X}Y$
  - $XY\bar{Z} + \bar{X}\bar{Y}Z$
- Verify using truth table for the following Boolean algebra.
  - $X + XY = X$
  - $\overline{X + Y} = \bar{X} \cdot \bar{Y}$

### ➤ Boolean Postulates:

- The fundamental laws of Boolean algebra are called as the postulates of Boolean algebra.
- These postulates for Boolean algebra originate from the three basic logic functions AND, OR and NOT.
- Properties of 0 and 1:**
  - If  $X \neq 0$  then  $X = 1$ , and If  $X \neq 1$  then  $X = 0$
  - OR relation ( Logical Addition)
    - $0 + 0 = 0$
    - $0 + 1 = 1$
    - $1 + 0 = 1$
    - $1 + 1 = 1$
  - AND relation ( Logical Multiplication)
    - $0 \cdot 0 = 0$
    - $0 \cdot 1 = 0$
    - $1 \cdot 0 = 0$
    - $1 \cdot 1 = 1$
  - Complement Rules
    - $\bar{0} = 1$
    - $\bar{1} = 0$

### ➤ Principle of Duality Theorem:

- This is very important principle used in Boolean algebra.
- Principle of Duality states that;
  - Changing each OR sign (+) to an AND sign (.)
  - Changing each AND sign (.) to an OR sign (+)
  - Replacing each 0 by 1 and each 1 by 0.
- The derived relation using duality principle is called dual of original expression.
- Example: Take postulate II, related to logical addition:
  - $0 + 0 = 0$
  - $0 + 1 = 1$
  - $1 + 0 = 1$
  - $1 + 1 = 1$
- Now working according to above relations, + is changed to . and 0's replaced by 1's
  - $1 \cdot 1 = 1$
  - $1 \cdot 0 = 0$
  - $0 \cdot 1 = 0$
  - $0 \cdot 0 = 0$
- which are nothing but same as that of postulate III related to logical multiplication.
- So 1, 2, 3, 4, are the duals of a, b, c, d.
- Example: Find the duals for the following Boolean Expression

| Sl No | Boolean Expression      | Duals                         |
|-------|-------------------------|-------------------------------|
| 1     | $X + 0 = X$             | $X \cdot 1 = X$               |
| 2     | $X + 1 = 1$             | $X \cdot 0 = 0$               |
| 3     | $X \cdot \bar{X} = 0$   | $X + \bar{X} = 1$             |
| 4     | $X \cdot (Y + Z)$       | $X + (Y \cdot Z)$             |
| 5     | $X + X \cdot Y = X + Y$ | $X \cdot (X + Y) = X \cdot Y$ |

➤ **Boolean Theorems:**

- Boolean Theorem can be proved by substituting all possible values of the variable that are 0 and 1.
- This technique of proving theorem is called **Proof by perfect induction.**

| Sl No                        | Theorem                 | Sl No                   | Theorem                                     |
|------------------------------|-------------------------|-------------------------|---------------------------------------------|
| <b>Properties of 0 and 1</b> |                         | <b>Associative Law</b>  |                                             |
| 1                            | $0 + X = X$             | 12                      | $X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$ |
| 2                            | $1 + X = 1$             | 13                      | $(X + Y) \cdot Z = X + (Y \cdot Z)$         |
| 3                            | $0 \cdot X = 0$         | <b>Distributive Law</b> |                                             |
| 4                            | $1 \cdot X = X$         | 14                      | $X \cdot (Y + Z) = X \cdot Y + X \cdot Z$   |
| <b>Indempotence Law</b>      |                         | 15                      | $X + Y \cdot Z = (X + Y) \cdot (X + Z)$     |
| 5                            | $X + X = X$             | <b>Absorption Law</b>   |                                             |
| 6                            | $X \cdot X = X$         | 16                      | $X + XY = X$                                |
| <b>Complementary Law</b>     |                         | 17                      | $X(X + Y) = X$                              |
| 7                            | $X + \bar{X} = 1$       | 18                      | $XY + X\bar{Y} = X$                         |
| 8                            | $X \cdot \bar{X} = 0$   | 19                      | $(X + Y)(X + \bar{Y}) = X$                  |
| <b>Involution Law</b>        |                         | 20                      | $X + \bar{X}Y = X + Y$                      |
| 9                            | $\bar{\bar{X}} = X$     | 21                      | $X(\bar{X} + Y) = XY$                       |
| <b>Commutative Law</b>       |                         |                         |                                             |
| 10                           | $X + Y = Y + X$         |                         |                                             |
| 11                           | $X \cdot Y = Y \cdot X$ |                         |                                             |

➤ **Theorem 1:  $0 + X = X$** 

| Proof: If $X = 0$<br>then LHS $= 0 + X$<br>$= 0 + 0$<br>$= 0$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= 0 + X$<br>$= 0 + 1$<br>$= 1$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>0</th><th>X</th><th>0+X</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> </table> | 0 | X | 0+X | 0 | 0 | 0 | 0 | 1 | 1 |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|---|---|---|
| 0                                                                               | X                                                                               | 0+X                                                                                                                                                                           |   |   |     |   |   |   |   |   |   |
| 0                                                                               | 0                                                                               | 0                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |
| 0                                                                               | 1                                                                               | 1                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |

➤ **Theorem 2:  $1 + X = 1$** 

| Proof: If $X = 0$<br>then LHS $= 1 + X$<br>$= 1 + 0$<br>$= 1$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= 1 + X$<br>$= 1 + 1$<br>$= 1$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>1</th><th>X</th><th>1+X</th></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </table> | 1 | X | 1+X | 1 | 0 | 1 | 1 | 1 | 1 |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|---|---|---|
| 1                                                                               | X                                                                               | 1+X                                                                                                                                                                           |   |   |     |   |   |   |   |   |   |
| 1                                                                               | 0                                                                               | 1                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |
| 1                                                                               | 1                                                                               | 1                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |

➤ **Theorem 3:**  $0 \cdot X = 0$ 

| Proof: If $X = 0$<br>then LHS $= 0 \cdot X$<br>$= 0 \cdot 0$<br>$= 0$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= 0 \cdot X$<br>$= 0 \cdot 1$<br>$= 0$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>0</th><th>X</th><th>0.X</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> </table> | 0 | X | 0.X | 0 | 0 | 0 | 0 | 1 | 0 |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|---|---|---|
| 0                                                                                       | X                                                                                       | 0.X                                                                                                                                                                           |   |   |     |   |   |   |   |   |   |
| 0                                                                                       | 0                                                                                       | 0                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |
| 0                                                                                       | 1                                                                                       | 0                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |

➤ **Theorem 4:**  $1 \cdot X = X$ 

| Proof: If $X = 0$<br>then LHS $= 1 \cdot X$<br>$= 1 \cdot 0$<br>$= 0$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= 1 \cdot X$<br>$= 1 \cdot 1$<br>$= 1$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>1</th><th>X</th><th>0.X</th></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </table> | 1 | X | 0.X | 1 | 0 | 0 | 1 | 1 | 1 |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|---|---|---|
| 1                                                                                       | X                                                                                       | 0.X                                                                                                                                                                           |   |   |     |   |   |   |   |   |   |
| 1                                                                                       | 0                                                                                       | 0                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |
| 1                                                                                       | 1                                                                                       | 1                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |

➤ **Idempotence Law:** “This law states that when a variable is combines with itself using OR or AND operator, the output is the same variable”.➤ **Theorem 5:**  $X + X = X$ 

| Proof: If $X = 0$<br>then LHS $= X + X$<br>$= 0 + 0$<br>$= 0$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= X + X$<br>$= 1 + 1$<br>$= 1$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>X</th><th>X</th><th>X+X</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </table> | X | X | X+X | 0 | 0 | 0 | 1 | 1 | 1 |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|---|---|---|
| X                                                                               | X                                                                               | X+X                                                                                                                                                                           |   |   |     |   |   |   |   |   |   |
| 0                                                                               | 0                                                                               | 0                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |
| 1                                                                               | 1                                                                               | 1                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |

➤ **Theorem 6:**  $X \cdot X = X$ 

| Proof: If $X = 0$<br>then LHS $= X \cdot X$<br>$= 0 \cdot 0$<br>$= 0$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= X \cdot X$<br>$= 1 \cdot 1$<br>$= 1$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>X</th><th>X</th><th>X.X</th></tr> <tr> <td>0</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>1</td></tr> </table> | X | X | X.X | 0 | 0 | 0 | 1 | 1 | 1 |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|-----|---|---|---|---|---|---|
| X                                                                                       | X                                                                                       | X.X                                                                                                                                                                           |   |   |     |   |   |   |   |   |   |
| 0                                                                                       | 0                                                                                       | 0                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |
| 1                                                                                       | 1                                                                                       | 1                                                                                                                                                                             |   |   |     |   |   |   |   |   |   |

➤ **Complementary Law:** “This law states that when a variable is And ed with its complement is equal to 0 and a variable is OR ed with its complement is equal to 1”.

➤ **Theorem 7:**  $X + \bar{X} = 1$ 

| Proof: If $X = 0$<br>then LHS $= X + \bar{X}$<br>$= 0 + 1$<br>$= 1$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= X + \bar{X}$<br>$= 1 + 0$<br>$= 1$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>X</th><th><math>\bar{X}</math></th><th><math>X + \bar{X}</math></th></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> </table> | X | $\bar{X}$ | $X + \bar{X}$ | 0 | 1 | 1 | 1 | 0 | 1 |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|---------------|---|---|---|---|---|---|
| X                                                                                     | $\bar{X}$                                                                             | $X + \bar{X}$                                                                                                                                                                                                         |   |           |               |   |   |   |   |   |   |
| 0                                                                                     | 1                                                                                     | 1                                                                                                                                                                                                                     |   |           |               |   |   |   |   |   |   |
| 1                                                                                     | 0                                                                                     | 1                                                                                                                                                                                                                     |   |           |               |   |   |   |   |   |   |

➤ **Theorem 8:**  $X \cdot \bar{X} = 0$ 

| Proof: If $X = 0$<br>then LHS $= X \cdot \bar{X}$<br>$= 0 \cdot 1$<br>$= 0$<br>$= \text{RHS}$ | Proof: If $X = 1$<br>then LHS $= X \cdot \bar{X}$<br>$= 1 \cdot 0$<br>$= 0$<br>$= \text{RHS}$ | Using Truth Table<br><table border="1"> <tr> <th>X</th><th><math>\bar{X}</math></th><th><math>X \cdot \bar{X}</math></th></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> </table> | X | $\bar{X}$ | $X \cdot \bar{X}$ | 0 | 1 | 0 | 1 | 0 | 0 |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------|-------------------|---|---|---|---|---|---|
| X                                                                                             | $\bar{X}$                                                                                     | $X \cdot \bar{X}$                                                                                                                                                                                                         |   |           |                   |   |   |   |   |   |   |
| 0                                                                                             | 1                                                                                             | 0                                                                                                                                                                                                                         |   |           |                   |   |   |   |   |   |   |
| 1                                                                                             | 0                                                                                             | 0                                                                                                                                                                                                                         |   |           |                   |   |   |   |   |   |   |

➤ **Involution Law:** “This law states that when a variable is inverted twice is equal to the original variable”.➤ **Theorem 9:**  $\bar{\bar{X}} = X$ 

| <p>Proof: If <math>X = 0</math>, then <math>\bar{X} = 1</math></p> <p>Take complement again, then <math>\bar{\bar{X}} = 0</math> i.e. <math>X</math></p> <p>If <math>X = 1</math>, then <math>\bar{X} = 0</math></p> <p>Take complement again, then <math>\bar{\bar{X}} = 1</math> i.e. <math>X</math></p> | <p>Using Truth Table</p> <table><tr><th>X</th><th><math>\bar{X}</math></th><th><math>\bar{\bar{X}}</math></th></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr></table> | X               | $\bar{X}$ | $\bar{\bar{X}}$ | 0 | 1 | 0 | 1 | 0 | 1 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|-----------------|---|---|---|---|---|---|
| X                                                                                                                                                                                                                                                                                                          | $\bar{X}$                                                                                                                                                                                                 | $\bar{\bar{X}}$ |           |                 |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                         | 0               |           |                 |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                          | 0                                                                                                                                                                                                         | 1               |           |                 |   |   |   |   |   |   |

➤ **Commutative Law:** “This law states that the order in which two variable are Or ed or AND ed make no difference”.➤ **Theorem 10:**  $X + Y = Y + X$ 

| Proof: If $Y = 0$<br><br>then LHS $= X + Y$<br><br>$= X + 0$<br><br>$= X$<br><br>RHS $= Y + X$<br><br>$= 0 + X$<br><br>$= X$<br><br>Therefore   LHS = RHS | Proof: If $Y = 1$<br><br>then LHS $= X + Y$<br><br>$= X + 1$<br><br>$= 1$<br><br>RHS $= Y + X$<br><br>$= 1 + X$<br><br>$= 1$<br><br>Therefore   LHS = RHS | Using Truth Table<br><table><tr><th>X</th><th>Y</th><th>X+Y</th><th>Y+X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> | X   | Y | X+Y | Y+X | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| X                                                                                                                                                         | Y                                                                                                                                                         | X+Y                                                                                                                                                                                                                                                                                           | Y+X |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                         | 0                                                                                                                                                         | 0                                                                                                                                                                                                                                                                                             | 0   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                         | 1                                                                                                                                                         | 1                                                                                                                                                                                                                                                                                             | 1   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                         | 0                                                                                                                                                         | 1                                                                                                                                                                                                                                                                                             | 1   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                         | 1                                                                                                                                                         | 1                                                                                                                                                                                                                                                                                             | 1   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

➤ **Theorem 11:**  $X \cdot Y = Y \cdot X$ 

| <p>Proof: If <math>Y = 0</math></p> <p>then LHS      <math>= X \cdot Y</math></p> <p>                 <math>= X \cdot 0</math></p> <p>                 <math>= 0</math></p> <p>RHS            <math>= Y \cdot X</math></p> <p>                 <math>= 0 \cdot X</math></p> <p>                 <math>= 0</math></p> <p>Therefore    LHS = RHS</p> | <p>Proof: If <math>Y = 1</math></p> <p>then LHS      <math>= X \cdot Y</math></p> <p>                 <math>= X \cdot 1</math></p> <p>                 <math>= X</math></p> <p>RHS            <math>= Y \cdot X</math></p> <p>                 <math>= 1 \cdot X</math></p> <p>                 <math>= X</math></p> <p>Therefore    LHS = RHS</p> | <p>Using Truth Table</p> <table><tr><th>X</th><th>Y</th><th>X.Y</th><th>Y.X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> | X   | Y | X.Y | Y.X | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| X                                                                                                                                                                                                                                                                                                                                                  | Y                                                                                                                                                                                                                                                                                                                                                  | X.Y                                                                                                                                                                                                                                                                                               | Y.X |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                 | 0   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                 | 0   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                 | 0   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                 | 1   |   |     |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

➤ **Associative Law:** “This law allows the removal of brackets from an expression and regrouping of the variables”.

➤ **Theorem 12:**  $X \cdot (Y \cdot Z) = (X \cdot Y) \cdot Z$ 

| <p>Proof: If <math>Y = 0</math></p> <p>LHS = <math>X.(Y.Z)</math></p> <p>= <math>X.(0.Z)</math></p> <p>= <math>X.0</math></p> <p>= <math>0</math></p> <p>RHS = <math>(X.Y).Z</math></p> <p>= <math>(X.0).Z</math></p> <p>= <math>0.Z</math></p> <p>= <math>0</math></p> <p>Therefore LHS = RHS</p> | <p>Proof: If <math>Y = 1</math></p> <p>LHS = <math>X.(Y.Z)</math></p> <p>= <math>X.(1.Z)</math></p> <p>= <math>X.Z</math></p> <p>= <math>XZ</math></p> <p>RHS = <math>(X.Y).Z</math></p> <p>= <math>(X.1).Z</math></p> <p>= <math>X.Z</math></p> <p>= <math>XZ</math></p> <p>Therefore LHS = RHS</p> | <p>Using Truth Table</p> <table><tr><th>X</th><th>Y</th><th>Z</th><th>XY</th><th>YZ</th><th>X.(Y.Z)</th><th>(X.Y).Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> | X  | Y  | Z       | XY      | YZ | X.(Y.Z) | (X.Y).Z | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---------|---------|----|---------|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| X                                                                                                                                                                                                                                                                                                  | Y                                                                                                                                                                                                                                                                                                    | Z                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | XY | YZ | X.(Y.Z) | (X.Y).Z |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 1  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0  | 0  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1  | 0  | 0       | 0       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1  | 1  | 1       | 1       |    |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

➤ **Theorem 13:**  $X + (Y + Z) = (X + Y) + Z$ 

| <p>Proof: If <math>Y = 0</math></p> <p>LHS = <math>X+(Y+Z)</math></p> <p><math>= X+(0+Z)</math></p> <p><math>= X+Z</math></p> <p>RHS = <math>(X+Y)+Z</math></p> <p><math>= (X+0)+Z</math></p> <p><math>= X+Z</math></p> <p>Therefore LHS = RHS</p> | <p>Proof: If <math>Y = 1</math></p> <p>LHS = <math>X+(Y+Z)</math></p> <p><math>= X+(1+Z)</math></p> <p><math>= X+1</math></p> <p><math>= 1</math></p> <p>RHS = <math>(X+Y)+Z</math></p> <p><math>= (X+1).Z</math></p> <p><math>= 1+Z</math></p> <p><math>= 1</math></p> <p>Therefore LHS = RHS</p> | <p>Using Truth Table</p> <table><tr><th>X</th><th>Y</th><th>Z</th><th>X+Y</th><th>Y+Z</th><th>X+(Y+Z)</th><th>(X+Y)+Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> | X   | Y   | Z       | X+Y     | Y+Z | X+(Y+Z) | (X+Y)+Z | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|---------|---------|-----|---------|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| X                                                                                                                                                                                                                                                  | Y                                                                                                                                                                                                                                                                                                  | Z                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | X+Y | Y+Z | X+(Y+Z) | (X+Y)+Z |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0   | 0   | 0       | 0       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0   | 1   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1   | 1   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1   | 1   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1   | 0   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1   | 1   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1   | 1   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1   | 1   | 1       | 1       |     |         |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |



➤ **Distributive Law:** “This law allows the multiplying or factoring out an expression”.

➤ **Theorem 14:**  $X.(Y+Z) = XY + XZ$

|                                                                                                                                    |                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Proof: If $X = 0$<br>$LHS = X.(Y+Z)$<br>$= 0.(Y+Z)$<br>$= 0$<br>$RHS = XY + XZ$<br>$= 0.Y + 0.Z$<br>$= 0$<br>Therefore $LHS = RHS$ | Proof: If $X = 1$<br>$LHS = X.(Y+Z)$<br>$= 1.(Y+Z)$<br>$= Y+Z$<br>$RHS = XY + XZ$<br>$= 1.Y + 1.Z$<br>$= Y+Z$<br>Therefore $LHS = RHS$ |
|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|

➤ **Theorem 15:**  $(X + Y) (X + Z) = X + YZ$

$$\begin{aligned}
 LHS: (X + Y) (X + Z) &= XX + XZ + XY + YZ \\
 &= X + XZ + XY + YZ \\
 &= X(1 + Z) + XY + YZ \\
 &= X + XY + YZ \\
 &= X(1 + Y) + YZ \\
 &= X + YZ \\
 &= RHS
 \end{aligned}$$

**Important**  
**2 Marks**

➤ **Absorption Law:** “This law enables a reduction of complicated expression to a simpler one by absorbing common terms”.

|                                                                                                                                                                                          |                                                                                                                                   |                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>16) <math>X+XY = X</math></b><br>$LHS = X + XY$<br>$= X(1 + Y)$<br>$= X$<br>$= RHS$                                                                                                   | <b>17) <math>X(X+Y) = X</math></b><br>$LHS = X(X+Y)$<br>$= XX + XY$<br>$= X + XY$<br>$= X(1+Y)$<br>$= X$                          | <b>18) <math>XY + X\bar{Y} = X</math></b><br>$LHS = XY + X\bar{Y}$<br>$= X(Y+\bar{Y})$<br>$= X.1$<br>$= X$<br>$= RHS$       |
| <b>19) <math>(X+Y)(X+\bar{Y}) = X</math></b><br>$LHS = (X+Y)(X+\bar{Y})$<br>$= XX + X\bar{Y} + XY + Y\bar{Y}$<br>$= X + X\bar{Y} + XY + 0$<br>$= X(1 + \bar{Y} + Y)$<br>$= X.1$<br>$= X$ | <b>20) <math>X + \bar{X}Y = X+Y</math></b><br>$LHS = X + \bar{X}Y$<br>$= (X + \bar{X})(X+Y)$<br>$= 1.(X+Y)$<br>$= X+Y$<br>$= RHS$ | <b>21) <math>X(\bar{X}+Y) = XY</math></b><br>$LHS = X(\bar{X}+Y)$<br>$= X.\bar{X} + X.Y$<br>$= 0 + XY$<br>$= XY$<br>$= RHS$ |

### ➤ DeMorgan's Theorem:

#### • DeMorgan's First Theorem:

- **Statement:** "When the OR sum of two variables is inverted, this is same as inverting each variable individually and then AND ing these inverted variables"
- This can be written as  $\overline{X + Y} = \bar{X} \cdot \bar{Y}$
- We can prove the DeMorgan's First theorem by using Truth Table is

| X | Y | $\bar{X}$ | $\bar{Y}$ | X+Y | $\overline{X + Y}$ | $\bar{X} \cdot \bar{Y}$ |
|---|---|-----------|-----------|-----|--------------------|-------------------------|
| 0 | 0 | 1         | 1         | 0   | 1                  | 1                       |
| 0 | 1 | 1         | 0         | 1   | 0                  | 0                       |
| 1 | 0 | 0         | 1         | 1   | 0                  | 0                       |
| 1 | 1 | 0         | 0         | 1   | 0                  | 0                       |



- Compare the column  $\overline{X + Y}$  and  $\bar{X} \cdot \bar{Y}$ . Both of these are identical. Hence the DeMorgan's first theorem is proved.

#### • DeMorgan's Second Theorem:

- **Statement:** "When the AND product of two variables is inverted, this is same as inverting each variable individually and then OR ing these inverted variables"
- This can be written as  $\overline{X \cdot Y} = \bar{X} + \bar{Y}$
- We can prove the DeMorgan's Second theorem by using Truth Table is:

| X | Y | $\bar{X}$ | $\bar{Y}$ | X.Y | $\overline{X \cdot Y}$ | $\bar{X} + \bar{Y}$ |
|---|---|-----------|-----------|-----|------------------------|---------------------|
| 0 | 0 | 1         | 1         | 0   | 1                      | 1                   |
| 0 | 1 | 1         | 0         | 0   | 1                      | 1                   |
| 1 | 0 | 0         | 1         | 0   | 1                      | 1                   |
| 1 | 1 | 0         | 0         | 1   | 0                      | 0                   |

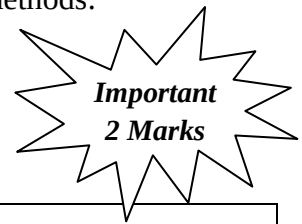
- Compare the column  $\overline{X \cdot Y}$  and  $\bar{X} + \bar{Y}$ . Both of these are identical. Hence the DeMorgan's Second theorem is proved.

#### • Application of DeMorgan's Theorem:

- It is used in simplification of Boolean expression.
- DeMorgan's law commonly apply to text searching using Boolean operators AND, OR and NOT.
- It is useful in the implementation of the basic gates operations with alternative gates.

➤ **Simplification of Boolean Expression:**

- Simplification of Boolean expression can be achieved by two popular methods:
  - Algebraic Manipulation
  - Karnaugh Maps
- **Algebraic Manipulation:**



|                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) $\bar{X} \bar{Y} Z + \bar{X} Y Z + X \bar{Y}$<br>$= \bar{X} \bar{Y} Z + \bar{X} Y Z + X \bar{Y}$<br>$= \bar{X} Z (\bar{Y} + Y) + X \bar{Y}$<br>$= \bar{X} Z (1) + X \bar{Y}$<br>$= \bar{X} Z + X \bar{Y}$ | 2) $XYZ + XYZW + XZ$<br>$= XYZ (1 + W) + XZ$<br>$= XYZ \cdot 1 + XZ$<br>$= XZ (Y + 1)$<br>$= XZ$                                                                                                                                                                   |
| 3) $Z(Y+Z)(X+Y+Z)$<br>$= (ZY + ZZ) (X+Y+Z)$<br>$= (ZY + Z) (X+Y+Z)$<br>$= Z(X+Y+Z)$ [Theorem 16 $X+XY=Z$ ]<br>$= ZX + ZY + ZZ$<br>$= ZX + ZY + Z$<br>$= Z(X+Y+1)$<br>$= Z(1)$<br>$= Z$                       | 4) $X + \bar{X}Y + \bar{Y} + (X + \bar{Y})\bar{X}Y$<br>$= X + \bar{X}Y + \bar{Y} + X\bar{X}Y + \bar{Y}\bar{X}Y$<br>$= X + \bar{X}Y + \bar{Y} + 0 + 0$<br>$= (X + \bar{X})(X + Y) + \bar{Y}$<br>$= 1(X + Y) + \bar{Y}$<br>$= X + Y + \bar{Y}$<br>$= X + 1$<br>$= 1$ |

➤ **Exercise Problems: Simplify using Algebraic Manipulation**

- 1)  $(\bar{A} + B) \cdot (A + B)$
- 2)  $AB + A\bar{B} + \bar{A}B$
- 3)  $B(A+C) + A\bar{B} + B\bar{C} + C$

➤ **Exercise Problems: Solving using DeMorgan's Theorem**

- 1)  $(\bar{A} + C) \cdot (B + D)$
- 2)  $A\bar{B} + C$

### ➤ Minterm:

- Minterm is a product of all the literal (with or without bar) within the logic system.
- **(OR)** A single variable or the logical product of several variables. The variables may or may not be complemented.
- A variable may appear either in its normal form (X) or in its complement form ( $\bar{X}$ )
- If a variable **value is 0 then its complemented** otherwise it is in its normal form.
- For example, if you have two variables X & Y, there are four possible combination can be formed with AND operation. Each of these four AND operations represents one of the Boolean expressions terms and is called a Minterm or a standard product.

| X | Y | Minterm          | Designation |
|---|---|------------------|-------------|
| 0 | 0 | $\bar{X}\bar{Y}$ | $m_0$       |
| 0 | 1 | $\bar{X}Y$       | $m_1$       |
| 1 | 0 | $X\bar{Y}$       | $m_2$       |
| 1 | 1 | $XY$             | $m_3$       |

- A symbol for each Minterm is also shown in the table and is of the form  $m_j$  where j denotes the decimal equivalent of the binary number of the Minterm designated.
- For example, the Minterm  $XY\bar{Z}$  whose combination is **1 1 0** can be written as  **$m_6$**  as decimal equivalent of **1 1 0** is **6**.
- A Boolean expression may be represented from a given truth table by forming a Minterm for each combination of the variables **which produces as 1 in the function, and then taking the OR (Logical Addition) of all those terms.**
- Assume the truth table

| X | Y | Z | Output   | Minterm                 | Designation |
|---|---|---|----------|-------------------------|-------------|
| 0 | 0 | 0 | 0        | $\bar{X}\bar{Y}\bar{Z}$ | $m_0$       |
| 0 | 0 | 1 | 0        | $\bar{X}\bar{Y}Z$       | $m_1$       |
| 0 | 1 | 0 | <b>1</b> | $\bar{X}Y\bar{Z}$       | $m_2$       |
| 0 | 1 | 1 | <b>1</b> | $\bar{X}YZ$             | $m_3$       |
| 1 | 0 | 0 | 0        | $X\bar{Y}\bar{Z}$       | $m_4$       |
| 1 | 0 | 1 | 0        | $X\bar{Y}Z$             | $m_5$       |
| 1 | 1 | 0 | <b>1</b> | $XY\bar{Z}$             | $m_6$       |
| 1 | 1 | 1 | 0        | $XYZ$                   | $m_7$       |

Minterm  
Results 1

- The Boolean function of truth table is obtained by OR ing (Add) three Minterm i.e. 010 ( $\bar{X}Y\bar{Z}$ ), 011 ( $\bar{X}YZ$ ), 110 ( $XY\bar{Z}$ ). Since each of these Minterm results is 1 (output).

$$f(X, Y, Z) = \bar{X}Y\bar{Z} + \bar{X}YZ + XY\bar{Z} = m_2 + m_3 + m_6$$

- The above Boolean function is the sum of three product terms. This type of expression is known as Sum of Product (SOP) expression.

$$f(X, Y, Z) = \sum (2, 3, 6)$$

- Where f is a Boolean function with three variables (X, Y, Z) and it can be read as function f is sum of 2<sup>nd</sup>, 3<sup>rd</sup>, and 6<sup>th</sup> Minterm.
- Sum of Product (SOP):** A Sum of product expression is a product term or several product terms logically added.

➤ **Find the Minterm designation for the following:**

1)  $X\bar{Y}\bar{Z}$

Binary Equivalent = 1 0 0

Decimal Equivalent =  $1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$   
 $= 4 + 0 + 0$   
 $= 4$

So,  $X\bar{Y}\bar{Z} = m_4$

2)  $A\bar{B}C\bar{D}$

Binary Equivalent = 1 0 1 0

Decimal Equivalent =  $1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$   
 $= 8 + 0 + 2 + 0$   
 $= 10$

So,  $A\bar{B}C\bar{D} = m_{10}$

- **What are the fundamental products for each of the input words; ABCD = 0010, ABCD = 110, ABCD = 1110. Write SOP expression.**

**Solution:** The SOP expression is:  $\bar{A}\bar{B}C\bar{D} + AB\bar{C} + ABC\bar{D}$

➤ **Maxterm:**

- Maxterm is a sum of all the literal (with or without bar) within the logic system.
- (OR)** A single variable or the logical sum of several variables. The variables may or may not be complemented.
- A variable may appear either in its normal form (X) or in its complement form ( $\bar{X}$ )
- If a variable value is 1 then its complemented otherwise it is in its normal form.
- For example, if you have two variables X & Y, there are four possible combination can be formed with OR operation. Each of these four OR operations represents one of the Boolean expressions

terms and is called a Minterm or a standard product.

| X | Y | Minterm             | Designation |
|---|---|---------------------|-------------|
| 0 | 0 | $X + Y$             | $M_0$       |
| 0 | 1 | $X + \bar{Y}$       | $M_1$       |
| 1 | 0 | $\bar{X} + Y$       | $M_2$       |
| 1 | 1 | $\bar{X} + \bar{Y}$ | $M_3$       |

- A symbol for each Maxterm is also shown in the table and is of the form  $M_j$  where  $j$  denotes the decimal equivalent of the binary number of the Maxterm designated.
- For example, the Maxterm  $X + Y + \bar{Z}$  whose combination is **0 0 1** can be written as  **$M_1$**  as decimal equivalent of **0 0 1** is **1**.
- A Boolean expression may be represented from a given truth table by forming a Maxterm for each combination of the variables **which produces as 0 in the function, and then taking the AND (Logical Multiplication) of all those terms.**
- Assume the truth table

| X | Y | Z | Output   | Minterm                       | Designation             |
|---|---|---|----------|-------------------------------|-------------------------|
| 0 | 0 | 0 | <b>0</b> | $X + Y + Z$                   | <b><math>M_0</math></b> |
| 0 | 0 | 1 | <b>0</b> | $X + Y + \bar{Z}$             | <b><math>M_1</math></b> |
| 0 | 1 | 0 | 1        | $X + \bar{Y} + Z$             | $M_2$                   |
| 0 | 1 | 1 | 1        | $X + \bar{Y} + \bar{Z}$       | $M_3$                   |
| 1 | 0 | 0 | <b>0</b> | $\bar{X} + Y + Z$             | <b><math>M_4</math></b> |
| 1 | 0 | 1 | <b>0</b> | $\bar{X} + Y + \bar{Z}$       | <b><math>M_5</math></b> |
| 1 | 1 | 0 | 1        | $\bar{X} + \bar{Y} + Z$       | $M_6$                   |
| 1 | 1 | 1 | <b>0</b> | $\bar{X} + \bar{Y} + \bar{Z}$ | <b><math>M_7</math></b> |

Maxterm  
Results 0

- The Boolean function of truth table is obtained by AND ing (Multiply) five Maxterm i.e. 000, 001, 100, 101, 111. Since each of these Maxterm results is 0 (output).

$$f(X, Y, Z) = (X + Y + Z)(X + Y + \bar{Z})(\bar{X} + Y + Z)(\bar{X} + Y + \bar{Z})(\bar{X} + \bar{Y} + \bar{Z}) = M_0.M_1.M_4.M_5.M_7$$

- The above Boolean function is the product of three sum terms. This type of expression is known as Product of Sum (POS) expression.

$$f(X, Y, Z) = \pi(0, 1, 4, 5, 7)$$

- Where  $f$  is a Boolean function with three variables ( $X, Y, Z$ ) and it can be read as function  $f$  is product of  $0^{\text{th}}$ ,  $1^{\text{st}}$ ,  $4^{\text{th}}$ ,  $5^{\text{th}}$  and  $7^{\text{th}}$  Maxterm.
- **Product of Sum (POS):** A product of sum expression is a sum term or several sum terms logically multiplied.